Electricity and Magnetism Isomorphic Questions

Electrostatics, Circuits, and Magnetism

Questions generated below. Multiple-choice answers in RED; incorrect answers in BLUE.

Q1. A negative point charge $-Q_1$ is fixed at the origin. A positive point charge $+Q_2$ is located on the negative x-axis. What is the direction of the Coulomb force acting on $+Q_2$, and what is the direction of the unit vector $\hat{r}$ in $\vec{F} = k\dfrac{(-Q_1)(+Q_2)}{r^2}\hat{r}$ when $\hat{r}$ is defined from the charge at the origin to $+Q_2$?

A. Force on $+Q_2$ points in the $-x$ direction; unit vector points in the $-x$ direction
This assumes the force must point in the same direction as $\hat{r}$. Because the charges are opposite, the force is attractive and points toward the origin, which is in the $+x$ direction from the location of $+Q_2$.

B. Force on $+Q_2$ points in the $+x$ direction; unit vector points in the $-x$ direction
The unit vector $\hat{r}$ is defined from the source charge at the origin to the field point where $+Q_2$ sits, so it points along the negative x direction. Since $q_1 q_2 < 0$, Coulomb's law gives a force opposite to $\hat{r}$, so the force on $+Q_2$ points in the positive x direction, toward the origin.

C. Force on $+Q_2$ points in the $+x$ direction; unit vector points in the $+x$ direction
This gets the attractive force direction right but reverses the definition of the separation unit vector. With the second charge on the negative x-axis, $\hat{r}$ from the origin to that charge must point in the $-x$ direction.

D. Force on $+Q_2$ points in the $-x$ direction; unit vector points in the $+x$ direction
This reverses both directions. Opposite charges attract, so the force on the positive charge must point toward the origin, not farther into the negative x region.

Answer: *B*

Q2. A negative point charge $-Q$ is fixed at the origin. A positive test charge $+q$ is placed on the negative y-axis. Which statement correctly describes the direction of the electric field at that location and the direction of the electric force on the positive charge?

A. Electric field upward; electric force upward
The electric field of a negative source points toward the source. At a point on the negative y-axis, that means the field points upward toward the origin, and because $\vec{F} = q\vec{E}$ for a positive test charge, the force is in the same upward direction.

B. Electric field upward; electric force downward
This treats the positive test charge as if it were negative. A positive charge experiences force in the same direction as the electric field, not opposite to it.

C. Electric field downward; electric force upward
This reverses the electric field direction. The field due to a negative source points toward the origin, so from a point below the origin it must point upward.

D. Electric field downward; electric force downward
This incorrectly makes both vectors point away from the source. A negative source creates a field toward itself, and a positive charge follows that field direction.

Answer: *A*

Q3. A point charge is placed at the center of a regular octahedral Gaussian surface. Which statement is correct?

A. The electric flux through every face is zero
This confuses zero field at some points with zero flux through a face. A closed surface enclosing a nonzero charge has nonzero total flux by Gauss's law.

B. The magnitude of the electric field is constant over each triangular face
Equal flux does not require uniform field over a face. Different points on one face are at different distances from the central charge, so the field magnitude varies.

C. The electric flux is the same through each face
Gauss's law gives the total flux as $\Phi_E = q/\varepsilon_0$ for the closed surface. Because all faces of a regular octahedron are equivalent by symmetry when the charge is at the center, the total flux is shared equally among them, so each face has the same flux.

D. None of the other statements is correct
This ignores the symmetry of the surface. The correct symmetric conclusion is that each face receives an equal share of the total electric flux.

Answer: *C*

Q4. An infinitely long straight wire carries a uniform negative linear charge density. A closed cylindrical Gaussian surface is placed coaxially around the wire. Which part or parts of the surface have nonzero electric flux?

A. Only the two flat circular end caps
This treats the field as if it were along the cylinder axis. For a line charge, the electric field is radial, so it is parallel to the end caps and gives zero flux there.

B. Both the curved surface and the end caps
This ignores the dot product in $\Phi = \int \vec{E} \cdot d\vec{A}$. The end-cap area vectors are axial while the field is radial, so the end-cap contribution is zero.

C. No part of the surface has nonzero flux
That would contradict Gauss's law for a surface enclosing charge. The net flux is nonzero and must pass through the curved side.

D. Only the curved side surface
For an infinite line charge, the electric field points radially inward or outward from the axis. On the curved side of a coaxial cylinder, $d\vec{A}$ is also radial, so $\vec{E} \cdot d\vec{A} \neq 0$, whereas on the end caps $d\vec{A}$ is along the axis and $\vec{E} \cdot d\vec{A} = 0$.

Answer: *D*

Q5. Two identical negative point charges, each of charge $-q$, are separated by a distance d in vacuum. Their separation is then increased to $3d$. How does the electric potential energy of the two-charge system change?

A. It increases
This would be true if the charges were opposite in sign. For like charges, the potential energy is positive and decreases as the separation increases.

B. It decreases
The electric potential energy of two point charges is $U = k\frac{q_1 q_2}{r}$. Here $q_1 q_2 = (+q^2)$ in magnitude because both charges are negative, so $U = kq^2/r$ is positive; increasing the separation from d to $3d$ changes it to $U = kq^2/(3d)$, which is smaller, so the energy decreases.

C. It becomes zero
Potential energy approaches zero only as the separation goes to infinity. At the finite distance $3d$, the potential energy is still positive and nonzero.

D. It remains unchanged
This ignores the $1/r$ dependence in $U = kq_1 q_2/r$. Changing the separation necessarily changes the electric potential energy.

Answer: *B*

Q6. A charged particle moves from point P to point Q in a uniform electric field. Which quantity determines the electric potential difference between P and Q?

A. The total distance traveled along the path
This treats potential difference as path-dependent. In electrostatics, the field is conservative, so only the endpoints matter.

B. The time required for the motion
Potential difference is a property of the field and positions, not of how long the particle takes to move. Time enters kinematics, not the definition of electric potential.

C. The component of displacement parallel to the electric field
In a uniform electric field, the potential difference is $\Delta V = -\vec{E}\cdot\Delta\vec{r}$. Therefore only the displacement component along the field contributes; any displacement perpendicular to the field does not change the electric potential.

D. The mass of the particle
Mass affects dynamics, not electric potential difference. The potential difference is set by the field and the locations of P and Q.

Answer: *C*

Q7. A parallel-plate capacitor has plate area A and separation d. If the plate area is reduced to $A/4$ while the plate separation remains unchanged, the capacitance:

A. Quadruples
This reverses the dependence on area. For parallel plates, larger area gives larger capacitance, so reducing area cannot make C increase.

B. Remains unchanged
This ignores the geometry dependence of a capacitor. The formula shows capacitance changes directly with plate area.

C. Is reduced by a factor of 2
This underestimates the effect of changing the area. Since $C \propto A$, reducing the area to one-fourth reduces the capacitance to one-fourth as well.

D. Is reduced by a factor of 4
For a parallel-plate capacitor, $C = \varepsilon_0 A/d$ or $C = \varepsilon A/d$ if a dielectric is present. Because capacitance is directly proportional to area, replacing A by $A/4$ makes the new capacitance $C/4$, so it is reduced by a factor of 4.

Answer: *D*

Q8. Two point charges are separated by 16.0 cm and have charges $+3.50\,\mu$C and $-3.50\,\mu$C, respectively. What is the magnitude of the electric field at the point midway between the charges? Use $k = 8.99 \times 10^9\,\mathrm{N \cdot m^2/C^2}$.

A. 0
This incorrectly assumes symmetry makes the fields cancel. At the midpoint between opposite charges, both field contributions point in the same direction, so they add rather than cancel.

B. 4.92×10^6 N/C
This is the field from only one charge at the midpoint. The net field must include both equal contributions, which point the same way.

C. 9.83×10^6 N/C
The midpoint is 0.080 m from each charge, so each charge produces $E = k|q|/r^2 = (8.99 \times 10^9)(3.50 \times 10^{-6})/(0.080)^2 \approx 4.92 \times 10^6$ N/C. Because the charges are opposite, the two fields at the midpoint point in the same direction, so the net magnitude is $2E \approx 9.83 \times 10^6$ N/C.

D. 1.97×10^7 N/C
This is too large, likely from doubling after already accounting for both charges or using half the distance incorrectly. The correct addition of the two equal fields gives 9.83×10^6 N/C.

Answer: *C*

Q9. A total charge of $+8.00\,\mu$C is uniformly distributed throughout an insulating sphere of radius 0.120 m. What is the magnitude and direction of the electric field at a point 0.180 m outside the surface of the sphere?

A. 2.22×10^6 N/C, radially outward
This likely uses 0.180 m as the distance from the center instead of from the surface. Outside the sphere, the relevant radius in $E = kQ/r^2$ is the full center-to-point distance $0.120 + 0.180 = 0.300$ m.

B. 7.99×10^5 N/C, radially outward
Because the point lies outside a spherically symmetric charge distribution, Gauss's law lets us treat the sphere as if all its charge were concentrated at the center. Using $r = 0.120 + 0.180 = 0.300$ m gives $E = kQ/r^2 = (8.99 \times 10^9)(8.00 \times 10^{-6})/(0.300)^2 \approx 7.99 \times 10^5$ N/C, and the field points radially outward because the total charge is positive.

C. 7.99×10^5 N/C, radially inward
The magnitude is correct, but the direction is not. A positive charge distribution produces an electric field directed away from the center, not toward it.

D. None of the above is correct
This overlooks that option B matches both the correct outside-field magnitude and the correct outward direction. The point is outside the sphere, so the point-charge formula applies.

Answer: *B*

Q10. A particle with charge $-2Q$ is placed at the center of an initially uncharged conducting hollow sphere. What charges are induced on the inner and outer surfaces of the sphere?

A. Inner: $-2Q$; outer: $+2Q$
This gives the inner surface the wrong sign. The inner surface must carry charge opposite to the enclosed charge so that the electric field inside the conducting material is zero.

B. Inner: $+Q$; outer: $-Q$
These amounts do not match the enclosed charge. To cancel the field inside the conductor, the inner surface must have total charge $+2Q$, not $+Q$.

C. Inner: 0; outer: $-2Q$
This ignores the required induction on the inner surface. A charge inside a cavity necessarily induces charge on the cavity wall so that the conducting material remains field-free.

D. Inner: $+2Q$; outer: $-2Q$
In electrostatic equilibrium, the electric field inside the conducting material must be zero, so a Gaussian surface just inside the conductor must enclose zero net charge. Therefore the inner surface must carry $+2Q$ to balance the central $-2Q$, and because the conductor as a whole started neutral, the outer surface must carry $-2Q$.

Answer: *D*

Q11. The electric potential along the x-axis is given by $V(x) = 2x^3 - 4x$, where V is in volts and x is in meters. What is the electric field at that point?

A. $\vec{E} = (6x^2 - 4)\hat{i}$
This is dV/dx, not $-dV/dx$. The common mistake is forgetting the minus sign in the relation between electric field and potential.

B. $\vec{E} = (4 - 6x^2)\hat{i}$
In one dimension, the electric field is $E_x = -\dfrac{dV}{dx}$. Since $\dfrac{dV}{dx} = 6x^2 - 4$, the field is $\vec{E} = -(6x^2 - 4)\hat{i} = (4 - 6x^2)\hat{i}$.

C. $\vec{E} = (2x^4 - 2x^2)\hat{i}$
This comes from an incorrect power manipulation. Differentiating a cubic potential should give a quadratic expression, not a quartic one.

D. $\vec{E} = (6x - 4)\hat{i}$
This differentiates x^3 incorrectly as if it were x^2. The derivative of $2x^3$ is $6x^2$, so the field must depend on x^2.

Answer: *B*

Q12. An 18 V battery is connected in series with a 4 μF capacitor, a 6 Ω resistor, an ammeter, and a switch that is initially open. A voltmeter is connected across the capacitor. At the instant the switch is closed, what are the ammeter and voltmeter readings, respectively?

A. **3.0 A, 0 V**
At $t = 0$, an initially uncharged capacitor behaves like a short circuit, so its voltage cannot jump instantly and $V_C(0) = 0$. The full battery voltage is across the resistor at that moment, so the current is $I(0) = V/R = 18/6 = 3.0$ A.

B. **0 A, 18 V**
This describes the long-time steady state after the capacitor is fully charged. It is not the behavior immediately after the switch is closed.

C. **0 A, 0 V**
This incorrectly assumes the capacitor blocks current instantly. Right after closing the switch, the uncharged capacitor allows maximum current in the charging circuit.

D. **3.0 A, 18 V**
The current value is correct for $t = 0$, but the capacitor voltage is not. A capacitor's voltage cannot change discontinuously from 0 to 18 V at the switching instant.

Answer: *A*

Q13. A battery charges a 2.2 mF capacitor through a 470 Ω resistor. What time is required for the capacitor's charge to reach 70% of its maximum value?

A. **0.36 s**
This is too short for reaching 70% of the final charge with this time constant. It likely comes from using the logarithm incorrectly or using the wrong fraction in the exponential.

B. **1.03 s**
This equals approximately one time constant RC, but at $t = RC$ the charge is about 63% of the maximum, not 70%. Reaching 70% requires a slightly longer time.

C. **1.24 s**
For charging, $q(t) = Q_{\max}\left(1 - e^{-t/RC}\right)$. Setting $q = 0.70Q_{\max}$ gives $0.70 = 1 - e^{-t/RC}$, so $e^{-t/RC} = 0.30$ and $t = -RC\ln(0.30)$; with $RC = (470)(2.2 \times 10^{-3}) = 1.034$ s, this yields $t \approx 1.24$ s.

D. **2.50 s**
This is much too long for only 70% charge. After this much time, the capacitor would be far closer to its final charge.

Answer: *C*

Q14. At one instant an electron has velocity components $v_x = -6.0 \times 10^5\,\text{m/s}$ and $v_z = 2.5 \times 10^5\,\text{m/s}$, so it is moving in the xz plane. A magnetic field of magnitude 0.40 T points in the positive z direction. At that instant, the magnetic force on the electron is:

A. $\mathbf{(3.84 \times 10^{-14}\,N)\hat{j}}$
This gets the magnitude right but forgets that the particle is negatively charged. The electron's charge reverses the direction of $\vec{v} \times \vec{B}$.

B. $\mathbf{(-1.60 \times 10^{-14}\,N)\hat{j}}$
This uses an incorrect magnitude, likely by including the wrong velocity component or making an arithmetic error. Only the component perpendicular to $\vec{B}$ contributes to the magnetic force.

C. $\mathbf{(-3.84 \times 10^{-14}\,N)\hat{k}}$
This puts the force along the field direction, which cannot happen for a magnetic force. Since $\vec{F} = q\vec{v} \times \vec{B}$, the force must be perpendicular to both $\vec{v}$ and $\vec{B}$.

D. $\mathbf{(-3.84 \times 10^{-14}\,N)\hat{j}}$
The magnetic force is $\vec{F} = q\vec{v} \times \vec{B}$. With $\vec{v} = (v_x, 0, v_z)$ and $\vec{B} = (0, 0, 0.40)$, only v_x contributes, giving $\vec{v} \times \vec{B} = (0, -v_x B, 0) = (0, 2.4 \times 10^5, 0)$; multiplying by $q = -1.60 \times 10^{-19}\,\text{C}$ gives $\vec{F} = (-3.84 \times 10^{-14}\,\text{N})\hat{j}$.

Answer: *D*